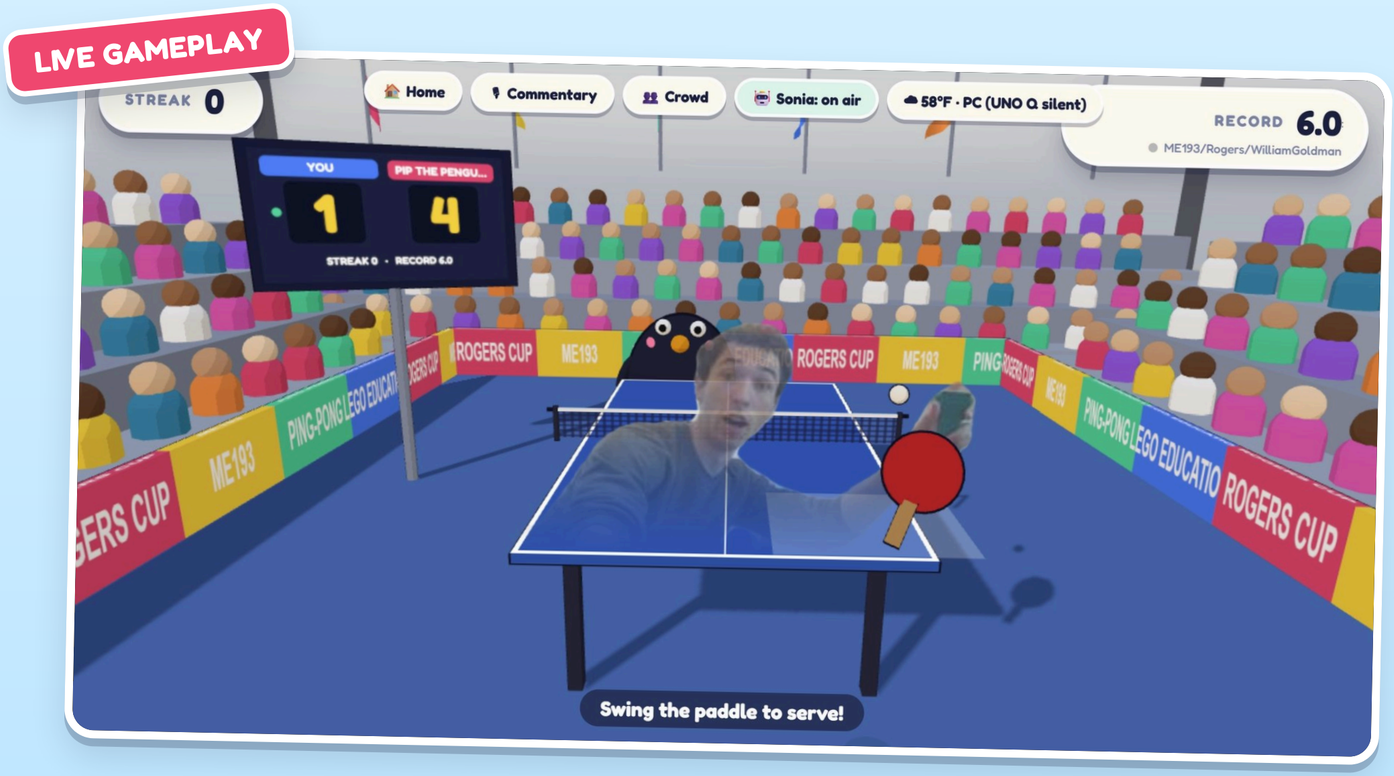
ME193 ROBOTICS · PROJECT 5

ROGERS CUP

Virtual table tennis where a LEGO motor is the paddle.

The webcam cuts you out of the room and puts you at a 3D table. Your pose tracks where the paddle is, the motor's gyro detects the swing, and two commentators call the match: one scripted, one written live by a local AI model.

- MediaPipe Pose
- LEGO Education BLE
- AprilTags
- Three.js / WebGL
- Qwen3-4B · OpenVINO
- Kokoro TTS
- Arduino UNO Q
- MQTT



↑ The player is cut out of the webcam image with a per-frame person mask. The red paddle follows their wrist; Pip the Penguin is the opponent.

William Goldman
Tufts University · ME193 Robotics · Fall 2026

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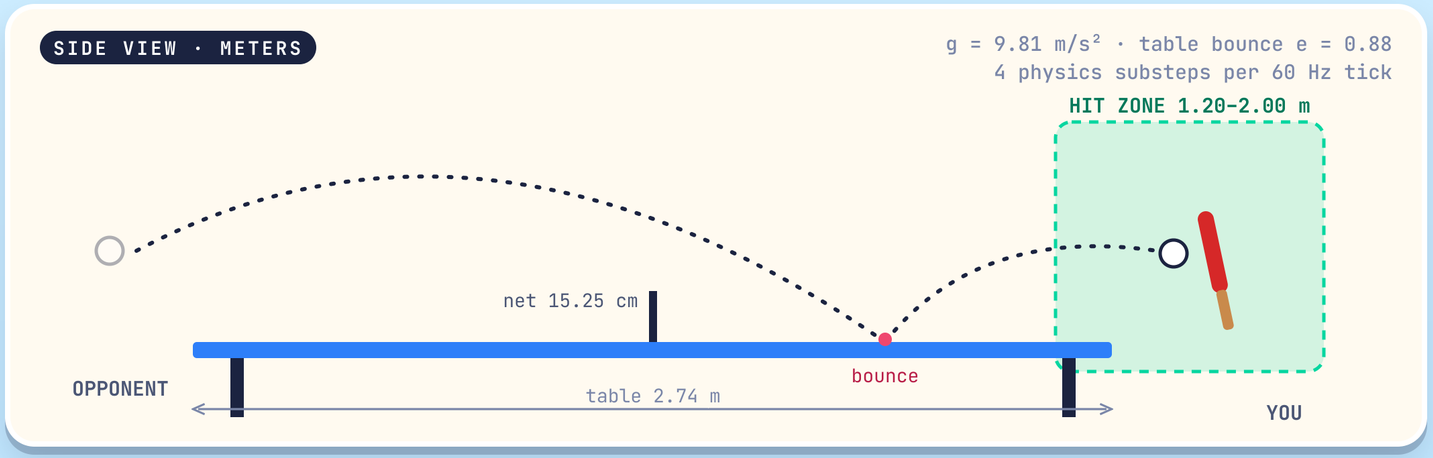
Sensors in, one game loop, many outputs

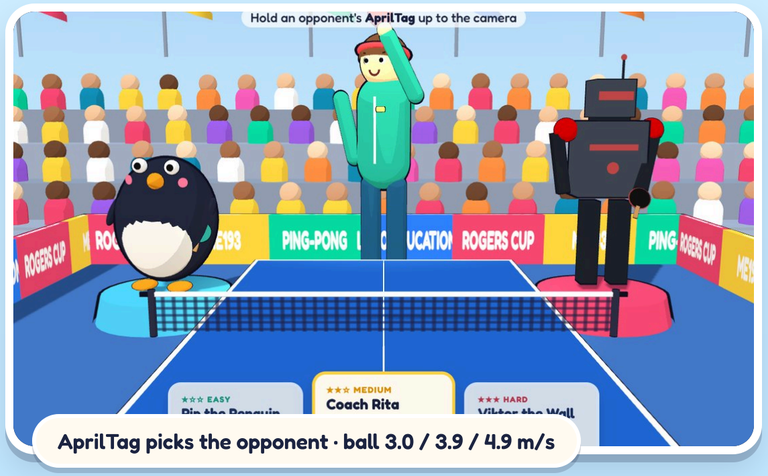
Python owns every rule and the physics. The browser only draws, so the game logic can be unit-tested without any hardware.



A hit needs two sensors to agree

Pose says **where** the paddle is. The IMU says **when** you swing. Either one alone can be faked; both together feel like real ping-pong.





AprilTag picks the opponent · ball 3.0 / 3.9 / 4.9 m/s

POSE · WHERE

Wrist relative to shoulders

Hand offset is measured in shoulder-widths, so distance from the camera doesn't matter.

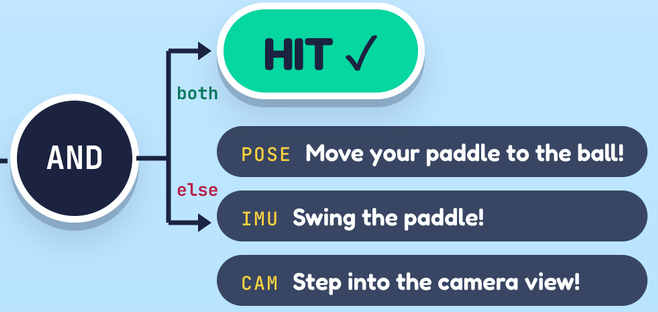
`paddle on ball: |Δx| ≤ 0.38 m, |Δy| ≤ 0.40 m`

IMU · WHEN

Gyro magnitude spike

Swing starts above the threshold and ends below 60% of it. Notifications raised from 100 to 30 ms.

`one swing per 300 ms · up to 0.25 s early counts`



Otherwise the miss names the signal that failed.

Two commentators, like a TV broadcast

Ray is instant because every line is pre-rendered. Sonia is written live by an open-source LLM on the laptop's own GPU: no API key, no cloud.

Ray PLAY-BY-PLAY · SCRIPTED

EVENT
Game event
point, deuce, streak, long rally, weather change

→

SCRIPT
Announcer picks a line
each line has an excitement level; hyped = higher, faster

→

≈1,200 CLIPS
Pre-rendered voice
Kokoro-82M, rendered once at setup. A score call joins two clips back to back.

Sonia COLOUR · LOCAL AI

MATCH STORY
What an analyst notices
momentum, repeated mistakes, comebacks

→

QWEN3-4B INT4
OpenVINO on Arc GPU
random focus per request: tip, crowd, joke…

→

FILTER
sanitize_line()
one sentence, ≤ 20 words, no score, no near-repeats

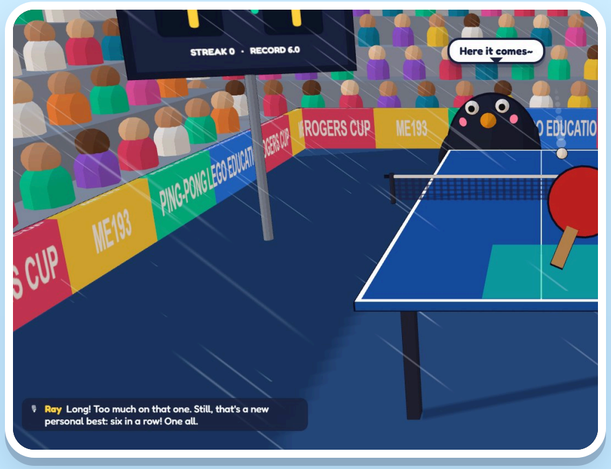
→

VOICE
Kokoro, 2 CPU threads
held until audio is ready; dropped once stale

1.3 s
per line, idle machine

2.5–4 s
per line, game running

Beat **Qwen3-1.7B** (faster but repetitive) and **Phi-4-mini** (slower, muddled) in a side-by-side comparison. The two share one speech queue: Ray's point calls cut in, Sonia waits her turn.



SONIA, IN TEST MATCHES

"The silence in the stands is louder than Viktor's next shot."

"Our player's feet are locked in place, but Viktor's spin is already on the other side of the net."

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The arena matches the weather outside

An app on the UNO Q publishes the live weather at Tufts. The sky, lighting, rain, snow and wind in the arena follow it, and the commentators talk about it.

UNO Q · ip-weather

Fetch every 5 min

Open-Meteo → scrolls "48F CLOUD" on the board's LED matrix

→

MQTT · RETAINED

Public broker

the game gets the latest reading the moment it subscribes; 60 s heartbeat

→

weather.py

Validate, then trust

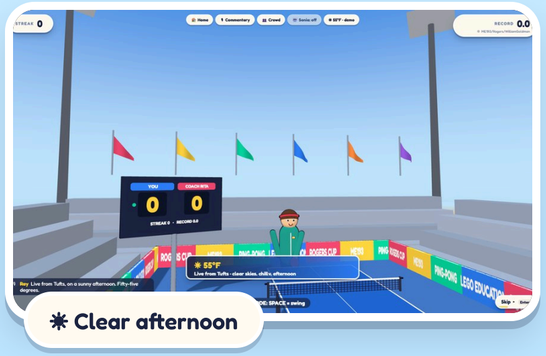
values clamped; the PC fetches it itself if the board is silent 3 min

→

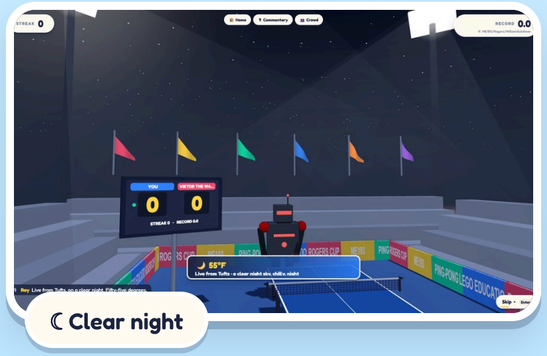
weather.js

Built once, re-tinted

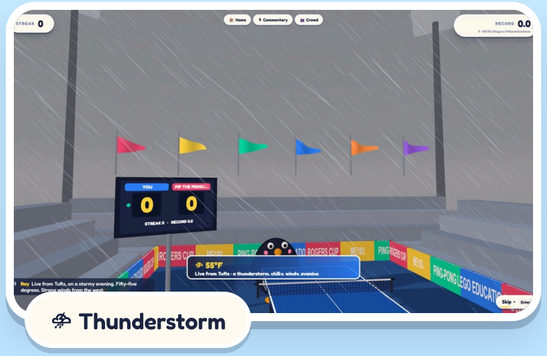
rain or snow is a single instanced draw; no shader recompiles mid-game



sun placed from the real sunrise and sunset



stars, moon phase, floodlights switch on



6,000 wind-slanted drops, lightning, wet ground



drifting flakes, the ground turns white

All four are screenshots of the running game. The lower-third card reads the same data the UNO Q publishes.

06 · PERFORMANCE

One laptop chip, three heavy jobs

Pose tracking (CPU), the 3D arena (GPU) and the LLM (GPU) share one Core Ultra 9 288V. Every fix below came from profiling the whole system second by second.



PROBLEM → FIX	BEFORE → AFTER
LLM generating mid-rally → gate it; a rally start cancels it within one token	camera 22 → 30 fps
Windows' 15.6 ms default timer → request 1 ms timers	game loop 32 → 60 Hz
Scheduler moving the game around → above-normal priority, no power throttling	worst tick gap 91 → 19 ms
Uncapped 120 fps at 2× → 60 fps at 1×, plus a quality governor	camera cost -8 fps → none

30 fps
camera, every rally second

60 Hz
game loop, page at 60 fps

35–40 ms
median frame age (max under 60 ms)

0
browser stalls

Three 100 s matches, AI live, synthetic 30 fps camera: the same numbers as with the AI turned off.